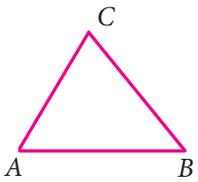
ABC is the required triangle.



Using trace copy superposition method, if we place the $\triangle ABC$ on the $\triangle PQR$ (Fig. 4.24) in such a way that the sides AB on PQ, AC on PR and the $\angle A$ on $\angle P$, then both the triangles ($\triangle ABC$ and $\triangle PQR$) matches exactly in size and shape. Here, we are using only two sides and one included angle to check the congruence of triangles.

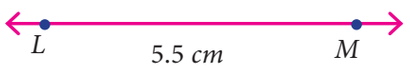
So we can say, if two sides and the included angle of a triangle are equal to the corresponding two sides and the included angle of another triangle, then the two triangles are congruent to each other.

This criterion is called Side – Angle – Side.

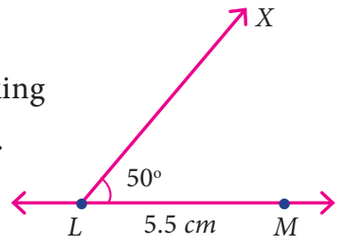
3. Angle – Side – Angle congruence criterion (ASA) - Two angles and the length of a side included by the angles are given.

Draw a triangle *LMN* given that $LM = 5.5\ cm$, $\angle M = 70°$ and $\angle L = 50°$.

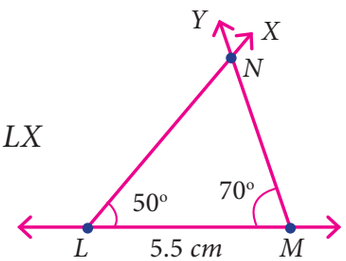
Step 1: Draw a line. Mark *L* and *M* on the line such that $LM = 5.5\ cm$.



Step 2: At *L*, draw a ray *LX* making an angle of 50° with *LM*.



Step 3: At *M*, draw another ray *MY* making an angle of 70° with *LM*. Mark the point of intersection of the rays *LX* and *MY* as *N*.



LMN is the required triangle.

Using trace copy superposition method, if we place the $\triangle LMN$ on the $\triangle PQR$ (Fig. 4.24) in such a way that the angles $\angle L$ on $\angle Q$, $\angle M$ on $\angle R$ and the side *LM* on *QR*, then both the triangles ($\triangle LMN$ and $\triangle PQR$) matches exactly in size and shape. Here, we used only two angles and one included side to check the congruence of triangles.

Therefore, we can say, if two angles and the included side of one triangle are congruent to the corresponding parts of another triangle, then the two triangles are congruent. This criterion is called Angle – Side – Angle criterion.

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